

12. The landform of 'Hanging Valley' is associated with which of the following geomorphic process?

- (a) Wind
- (b) Glacier
- (c) Sea Waves
- (d) Running Water

Uttarakhand P.C.S. (Pre) 2024

Ans. (b)

A Hanging Valley is a geomorphic feature formed primarily by Glacial Erosion. It occurs when a smaller tributary glacier joins a larger valley glacier, but due to the greater erosional power of the main glacier, the tributary valley is left at a higher elevation after the glaciers retreat.

This results in a steep drop or waterfall at the junction of the two Valleys. Famous examples include. Yosemite valley in the USA.

13. Which of the following landforms is found in Karst region?

- (a) Pot hole
- (b) Sink hole
- (c) Blow hole
- (d) Playa

Jharkhand P.C.S. (Mains) 2016

Ans. (b)

A doline (or sinkhole, as it is more commonly called in North America) is a natural enclosed depression found in karst landscapes. It is the most common landform in karst areas. Significantly, they are described as small to medium sized closed depressions, ranging from meters to tens of metres in both diameter and depth.

The Arid Regions/ Deserts

*Desert is a barren area of landscape where little precipitation occurs and consequently, living conditions are hostile for plants and animals. About 1/3rd of the land surface of the world is arid or semiarid. Regions receiving precipitation of less than 25 cm annually are categorized as deserts. Deserts can be classified by the amount of precipitation, by the temperature that prevails, by the causes of desertification or by their geographical location. ***Tropical hot deserts** are generally found on the western side of a continent between 15° to 30° north or south of the equator. Most of the major deserts of the world lie in areas crossed by the trade winds. *According to German scientist Dr. Gerhard Knies, the amount of energy that is produced by the world's deserts in 6 hours is more than the amount of energy consumed by the entire humankind in a year. Hence deserts are a very prominent source of renewable energy. ***The Sahara**

Desert is the largest non-polar desert in the world. The desert comprises much of North Africa, excluding the fertile regions on the Mediterranean Sea coast. It occupies a total area of 3600000 sq. miles. The Sahara covers large parts of Algeria, Chad, Egypt, Libya, Mali, Mauritania, Morocco, Niger, Western Sahara, Sudan and Tunisia. The main phenomena behind the formation of African and Euro-Asian deserts are –

1. Their proximity to cold ocean currents which run along the west coast. These cold ocean currents cool the air and make it harder for the air to hold moisture.
2. Most of these deserts are located far away from the moisture-laden trade winds.

***The Atacama** is a desert plateau in South America on the Pacific Coast, west of Andes Mountains. It is the driest desert in the world.

*The Atacama Desert is rich in metallic mineral resources such as copper, gold, silver and iron, as well as non-metallic minerals including important deposits of boron, lithium, sodium nitrate and potassium salts. It is the world's largest supplier of sodium nitrate. The Atacama Desert occupies the narrow coast of northern Chile and southern Peru.

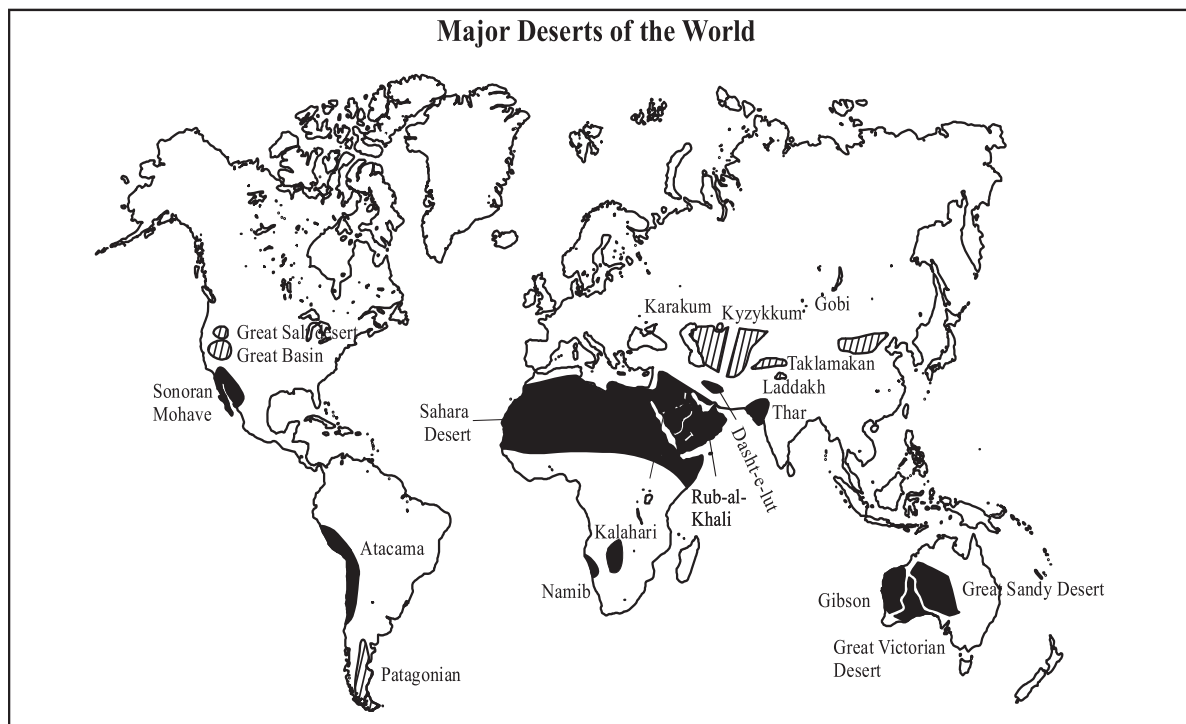
***The Kalahari Desert** is located in the Southern Hemisphere. The Kalahari Desert is a large semi-arid sandy savanna in Southern Africa, extending for 9,30,000 square kilometers covering much of Botswana, parts of Namibia and regions of South Africa.

***The Dasht-e-Lut** is a large salt desert located in the provinces of Kerman and Sistan and Baluchistan, Iran. According to NASA, it is the world's hottest desert. The Aqua satellite of NASA, in the year 2005 measured the surface temperature of this desert and found it to be as high as 70.7°C, making it the hottest desert (based on surface temperature) in the world.

***The Thar Desert** also known as the Great Indian Desert is a large arid region in the northwestern part of the Indian subcontinent and forms a natural boundary between India and Pakistan. It is the largest desert in South Asia. The Thar Desert is the most densely populated desert in the world, with a population density of 83 people per/km².

***The Taklamakan Desert** is a desert in southwestern Xinjiang Province in north west China. It is bounded by the Kunlun mountains to the south, the Pamir mountains to the west Tian Shan to the north, and the Gobi Desert in the east.

***The Sonoran Desert** occurs primarily in Mexico. More than two thirds of its total area is in Baja California, Baja California Sur, and the state of Sonora. It covers large parts of



the southwestern United States in Arizona and California. The western portion of the United States – Mexico border passes through the Sonoran Desert.

***The Karakum Desert**, is a desert in Central Asia. Its name means "black sand" in Turkic languages, in reference to the dark soil that lies beneath the sandy surface of much of the desert. It occupies much of the area of Turkmenistan and a smaller part of Kazakhstan.

***The Karakum Desert** is home to the Darwaza Gas (Crater). The area has significant oil and natural gas deposits.

***The Gibson Desert** is a large desert that covers a large dry area in the State of Western Australia. It is the 5th largest desert in Australia, after the Great Victoria, Great Sandy, Tanami and Simpson desert.

***Al Nafud** is a desert in Northern Saudi Arabia. The discovery of an 85,000 year old fossilized human finger in Al Nafud in 2016 by Dr. Mathieu Duval of Griffith University provided the earliest evidence of modern humans outside Africa and the Levant.

***The Rub'al Khali Desert** is located in Saudi Arabia, Oman, United Arab Emirates and Yemen are the largest contiguous sand desert in the world. Famous oil fields, Shaybah and Ghawar are located in this desert. ***The Kyzylkum Desert** is divided between Kazakhstan, Uzbekistan and Turkmenistan. The desert is well known for its deposits of gold, uranium, copper, aluminium and silver, natural gas and oil. ***Dasht-e-Kavir** is a

large desert located in Iran. ***The Gobi Desert** is a cold desert found in a temperate region. It covers parts of Northern China and of Southern Mongolia. The Gobi Desert is notable in history as part of the great Mongol Empire and as the location of several important cities along the Silk Road. ***Patagonia Desert** is a cold desert and also the largest desert in Argentina. It is located primarily in Argentina with small parts in Chile and is bounded by the Andes, to its west and the Atlantic Ocean to its east.

1. Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R). Choose the correct answer from the codes given below:

Assertion (A) : Most of the hot deserts of the world are located on the Western margins of continents between 15°-30° North and South latitudes.

Reason (R) : Trade winds lose their moisture and humidity as it move westward.

Codes :

- (a) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (c) (A) is true but (R) is false
- (d) (A) is false but (R) is true

U.P. B.E.O. (Pre) 2019

Ans. (a)

Most of the hot deserts of the world are located on the western margins of continents between 15° - 30° North and South Latitudes. Sahara and Kalahari deserts in Africa and Atacama in South America are examples.

Trade winds towards this region reach from the Northern Hemisphere and Southern Hemispheres. Thus, upon reaching the western part of the continent, they turn dry. This causes very little rainfall in this region. It is an important factor for desert formation in this region. The reason, therefore, is the correct interpretation of the statement. Hence, option (a) is correct.

2. **Statement (A) : Deserts can be effective sources for the production of perennial energy.**

Reason (R) : As much energy is consumed by mankind in one year, Desert can receive the energy is more than that in only 6 hour from the Sun.

Choose the correct answer from the following code –

- (a) (A) and (R), both are correct and (R) is the correct explanation of (A).
- (b) (A) and (R), both are correct, but (R) is not the correct explanation of (A).
- (c) (A) is correct, but (R) is false.
- (d) (A) is false but (R) is correct.

U.P.P.C.S. (Pre) 2013

Ans. (a)

Gerhard Knies, a German particle physicist, arrived at the remarkable conclusion that in just six hours, the world's deserts receive more energy from the sun than humans consume in a year. In this way, deserts can be a source of perennial energy. So, both the statement and reason are correct and the reason is the correct explanation of the statement.

3. **Desert is defined as the area where –**

- (a) Temperature remains more than 42° C
- (b) No plants are found
- (c) Annual rain occurs less than 25 cm
- (d) Prominence of sand barrows

U.P. Lower Sub. (Pre) 2009

Ans. (c)

Desert is a region where annual precipitation is less than 25 cm.

4. **The probability of having a desert on the earth is more-**

- (a) Near Zero degree latitude
- (b) Near 23° latitude

- (c) Near 50° latitude
- (d) Near 70° latitude

41st B.P.S.C. (Pre) 1996

Ans. (b)

The extension of hot dry desert climate on the earth is found between 15-30° latitudes. There are four major geographical conditions for the formation of deserts on the earth :

- (1) Presence of high air pressure around 30° latitudes. Here the winds keep descending from top to bottom, as a result of which the weather remains clear throughout the year. Sahara and Australian deserts come under this phenomenon.
- (2) The deserts of the western coast of the continents are found between 20-30° latitudes. Probably the reason for this is the eastern direction of the winds. Trade winds enter from the eastern part of the continents. By the time they reach the western part, they become dry. California (North America) W. Sahara (North Africa), Atacama (South America) and Namibia (South - West Africa) are similar deserts.
- (3) Among the desert located in the rain shadows region of the mountains, North American desert (Death Valley), Patagonia desert (Argentina) and Peruvian desert are prominent.
- (4) The deserts located in the middle of the continents, which are located away from the humid winds, include the Australian desert and the Gobi (China and Mongolia).

5. **The problem of desertification is least in which of the following continents?**

- (a) Europe
- (b) Australia
- (c) Asia
- (d) Africa
- (e) Question not attempted

Raj. P.C.S. (Pre) 2023

Ans. (a)

Europe is the only continent without a desert. Significantly, desertification is defined by the United Nations Convention to Combat Desertification (UNCCD) as "land degradation in arid, semi-arid and dry sub-humid areas (dryland) resulting from various factors, including climatic variations and human activities. Climatic variations include the decrease in precipitation, drought or soil and moisture loss on a global level. Notably, human activities include overgrazing, deforestation, removal of the natural vegetation cover and unsustainable agricultural activities in vulnerable ecosystems.

6. The biggest desert in the world is :

- (a) Kalahari (b) Gobi
(c) Sahara (d) Thar

M.P.P.C.S. (Pre) 2010

U.P.P.C.S. (Pre) 1994

Ans. (c)

The biggest (non-polar) desert in the world is the Sahara desert. It's situated in North Africa and has an area measured at roughly 3.6 million square miles.

7. Which of the following is the biggest Desert in the world—

- (a) Kalahari (b) Gobi
(c) Sahara (d) Thar

Uttarakhand Lower Sub. (Pre) 2010

Ans. (c)

See the explanation of the above question.

8. The most populated desert in the world is –

- (a) Sahara (b) Gobi
(c) Thar (d) Kalahari

U.P.P.C.S.(Pre) 2012

Ans. (c)

The Thar desert is the most densely populated desert in the world with a population density of 83 people per square km.

9. Takla Makan desert lies in –

- (a) Afghanistan (b) China
(c) Kazakhstan (d) Mongolia

U.P.P.C.S. (Spl) (Mains) 2008

Ans. (b)

Takla Makan Desert is situated in the South Western Xinjiang Province of North West China.

10. Which of the following is not correctly matched?

- | Desert | Country |
|----------------|----------------------------|
| (a) Sonoran | - United States of America |
| (b) Taklamakan | - China |
| (c) Karakum | - Turkmenistan |
| (d) Gibson | - Brazil |

U.P.P.C.S. (Mains) 2016

U.P. P.C.S. (Pre) 2018

Ans. (d)

Gibson desert, an interim Australian bioregion, is a large desert that covers a large dry area in the State of Western Australia not in Brazil. Other options are correctly matched.

11. Kara Kum desert is situated in which of the following countries?

- (a) Tajikistan (b) Kazakhstan
(c) Turkmenistan (d) Kirgistan

U.P.P.C.S. (Pre) 2022

Ans. (c) & (b)

Karakum Desert, also spelled Kara-Kum, is a great sandy region in Central Asia. It occupies mainly Turkmenistan and some part of Kara-Kum also situated in Kazakhstan.

12. Dasht-e- Lut is located in –

- (a) Iran (b) Libya
(c) Kenya (d) Nigeria

U.P.P.C.S. (Mains) 2015

Ans. (a)

Dasht-e-Lut is a large salt desert in Kerman and Sistan and Baluchestan provinces of Iran. NASA's tests have revealed that it is the world's warmest desert. Its temperature was measured at 70.7°C by NASA's Aqua satellite in 2005.

13. In which country is the Gobi desert located?

- (a) Mexico (b) Somalia
(c) Mongolia (d) Egypt

Chhattisgarh P.C.S. (Pre) 2008

U.P.P.C.S. (Mains) 1993

38th B.P.S.C. (Pre) 1992

Ans. (c)

Gobi Desert stretches into northern China and southern Mongolia. This desert is surrounded by mainly Alashan, Tienshan and Altai mountains.

14. Consider the following deserts and arrange them in increasing order of their size (from smallest to largest)

(I) Great Sandy desert

(II) Arabian desert

(III) Sahara desert

(IV) Gobi desert

Select the correct answer using the code given below :

- (a) II, I, III, IV (b) I, IV, II, III
(c) I, II, IV, III (d) II, I, IV, III

U.P.P.C.S. (Pre) 2023

Ans. (b)

The Great Sandy Desert is a desert in the northwest of Australia, mostly in Western Australia. It is the second-largest desert in Australia (after the Great Victoria Desert). It covers about 284,993 square kilometres (110,036 sq mi). **Arabian Desert** is a great desert region of extreme southwestern Asia that occupies almost the entire Arabian Peninsula. It is the largest desert area on the continent—covering an area of about 900,000 square miles (2,300,000 square km)—and the second largest on Earth, surpassed in size only by the Sahara, in northern Africa. **Sahara Desert** is the largest desert in the world, filling nearly all of northern Africa, it measures approximately 3,000 miles (4,800 km) from east to west and between 800 and 1,200 miles from north to south and has a total area of some 3,320,000 square miles (8,600,000 square km). **Gobi**, also called Gobi Desert, is a great desert and semi-desert region of Central Asia. The Gobi stretches across huge portions of both Mongolia and China. The Gobi occupies a vast arc of land 1,000 miles (1,600 km) long and 300 to 600 miles (500 to 1,000 km) wide, with an estimated area of 500,000 square miles (1,300,000 square km). Hence, option (b) is the correct answer.

15. The largest desert of South Asia is :

- (a) Atacama (b) Colorado
(c) Kalahari (d) Thar

U.P.P.C.S. (Pre) 1996

Ans. (d)

Thar desert which is also known as 'Great Indian Desert'. It is the largest desert of South Asia. It stretches across northwestern India to southeast Pakistan. The location of the remaining deserts is as follows :

Atacama Desert	- Chile and Peru
Colorado Desert	- U.S.A. and Mexico
Kalahari Desert	- Botswana, Namibia and South Africa

16. Which one of the following statements is not true regarding the Atacama ?

- (a) It is a desert in South America.
(b) It is World's driest desert.
(c) It lies in Southern Chile.
(d) It has rich reserves of nitrates.

U.P.P.C.S. (Mains) 2006

Ans. (c)

The Atacama Desert, located in South America, is the driest place on the Earth. There are huge deposits of sodium nitrate, copper and other minerals. This desert stretches along northern Chile and southern Peru and not in southern Chile.

17. The Atacama is located in –

- (a) Asia (b) South America
(c) Africa (d) North America

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

See the explanation of the above question.

18. The Driest place on earth is –

- (a) Arabian desert (b) Atacama desert
(c) Gobi desert (d) Thar desert

U.P. Lower Sub. (Pre) 2009

Ans. (b)

See the explanation of the above question.

19. Which one of the following is a temperate desert ?

- (a) Arabian Desert (b) Atacama Desert
(c) Kalahari Desert (d) Patagonian Desert

U.P.P.C.S. (Mains) 2011

Ans. (d)

Temperate desert is found between 30° to 60° latitudes. Deserts which come under the category of temperate deserts, are Gobi Desert, Ladakh Desert and Patagonia Desert.

20. Consider the following statements :-

- (1) Tropical deserts in the world are found in the trade wind bands of western ends of Continents-
(2) In India, Eastern Himalayan region receives more rain from North-eastern winds.

Which among the above statements is/are/ correct ?

Code :

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

I.A.S. (Pre) 2009

Ans. (a)

Tropical hot deserts are often found in the region of trade wind band between 15° to 30° South or North of the Equator on the western side of continents. Therefore statement 1 is correct. In India, Eastern Himalayan region receives more rain from the southwest winds coming from the Bay of Bengal. Therefore, statement 2 is wrong.

21. The Kalahari Desert lies in

1. Botswana
2. Namibia
3. Zaire
4. Zambia

Select the correct answer from the code given below –

Code :

- (a) 1 only
- (b) 2 only
- (c) 1 and 2
- (d) 3 and 4

U.P.P.C.S. (GIC) 2010

U.P.P.C.S. (Mains) 2008

Ans. (c)

The Kalahari is a large desert spreading over 930000 sq km. in the interior of Southern Africa. It mainly occupies Botswana, parts of Namibia and South Africa. Since, South Africa is not mentioned in the options, therefore, the correct answer will be option (c) (Botswana and Namibia).

22. Which of the following deserts is located in the Southern Hemisphere?

- (a) Kalahari
- (b) Gobi
- (c) Chihuahuan
- (d) Sahara

U.P.P.C.S. (Pre) (Re-Exam) 2015

Ans. (a)

The Kalahari Desert is located in the Southern Hemisphere. It spreads over 930000 square km across (mainly) Botswana, Namibia and South Africa. The name of the desert comes from the Tswana word Kgala, which means "the great thirst" or "a waterless place".

23. What could be the main reason/reasons for the formation of African and Eurasian desert belt?

1. It is located in the sub-tropical high-pressure cells.
2. It is under the influence of warm ocean currents.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

I.A.S. (Pre) 2011

Ans. (a)

Most desert regions are the result of large climatic patterns. The sub-tropical high-pressure belts are adiabatic in nature hence they bring little rainfall to the African and Eurasian desert regions. Warm ocean currents are not a necessary condition for the desert belt. For instance the, Gobi cold desert doesn't come under the influence of any warm currents.

24. Desert dwelling animals are called :

- (a) Xerocoles animals
- (b) Arboreal animals
- (c) Fossorial animals
- (d) Terrestrial animals

U.P.P.C.S. (Pre) 2023

Ans. (a)

Desert dwelling animals are called Xerocoles animals. Fossorial animals are those adapted to digging and life underground, while arboreal animals are those adapted to life in the trees and terrestrial animals are adapted to life on land.

25. The Great Victorian Desert is located at –

- (a) Australia
- (b) India
- (c) Egypt
- (d) North Africa
- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (a)

The Great Victorian Desert is located in Western Australia and South Australia. It is the largest desert in Australia. It is spread over 348,750 sq. km of area.

26. Match List-I with List-II and select the correct answer by using the codes given below :

List - I (Desert)	List-II (Location)
A. Gobi	i. Southern Africa
B. Patagonia	ii. Asia
C. Victoria	iii. South America
D. Kalahari	iv. Australia

Codes :

- | | | | |
|---------|-----|-----|----|
| A | B | C | D |
| (a) ii | iii | iv | i |
| (b) iv | ii | iii | i |
| (c) ii | i | iii | iv |
| (d) iii | iv | i | ii |

Raj. P.C.S. (Pre) 2024

Ans. (a)

The correct matching is -

Desert	Location
Gobi	Asia
Patagonia	South America
Victoria	Australia
Kalahari	Southern Africa